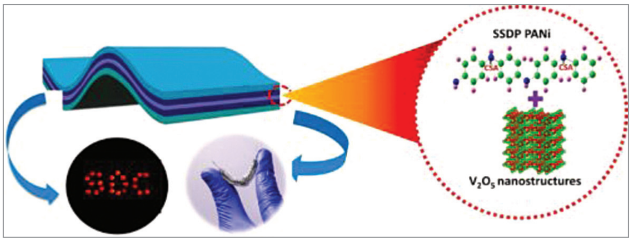

The emergence of a hybrid electrode-based, flexible, symmetric supercapacitor exhibiting electrochemical characteristics, cycle stability, and a high energy density presents a prospect for addressing future energy storage needs (Credit: https://www.pib.gov.in)

conductive hybrid electrode with a synergistic effect, outperforming traditional supercapacitors. The resulting supercapacitor device showcases superior electrochemical attributes, positioning it as a leading technology in energy storage.

Enhancing sustainable food, energy, and land productivity with agrivoltaics

Researchers from Texas A&M AgriLife are actively engaged in agrivoltaics, combining solar power arrays with crop and livestock farming to optimise land use, increase profitability, and address environmental challenges. This approach

Texas A&M AgriLife scientists are contributing to the growing body of research surrounding agrivoltaics, which combines agriculture and solar power production (Credit: Basia Latawiec)

improves water efficiency and mitigates heat in agriculture, potentially increasing crop yields and providing shade for livestock. Agrivoltaics can boost carbon sequestration and biodiversity, but challenges such as high initial costs and economic viability require further interdisciplinary research.

All-analogue chip combining electronic and light computing

Scientists at Tsinghua University have developed an all-analogue chip (ACCEL) fusing diffractive optical analogue computing (OAC) and electronic analogue computing (EAC), outperforming traditional digital processors in computer vision processing. ACCEL achieves high energy efficiency and processing speed by using optical computing for initial feature extraction and processing ensuing photocurrents within the same chip. It demonstrates high accuracy on various tasks, making it suitable for applications such as wearable technology, autonomous vehicles, and industrial checks.

Smart threads for next-generation fabrics

MIT researchers have introduced FibeRobo, a programmable fibre that contracts with temperature changes, allowing for mass production and programmable clothing features. Made from liquid crystal elastomer (LCE), this fibre is compatible

The affordable FibeRobo, fitting seamlessly with current textile methods, holds potential for adaptive sportswear and compression clothing (Credit: Courtesy of the researchers)

with standard textile methods, providing a low-cost solution for applications like sports bras or compression jackets for dogs. The team aims to make the fibre recyclable or biodegradable while simplifying its production process.

Electric vehicle usage and emissions impact study

A team from George Washington University and the National Renewable Energy Laboratory scrutinised odometer readings from nearly 12.9 million used cars and 11.9 million used SUVs over the period from 2016 to 2022. The research reveals that battery electric vehicles (BEVs) are driven less—around 7,250km (4,500 miles) less annually—than petrol (gasoline) vehicles. Electric cars average 11,530km (7,165 miles) per year compared to 18,735km (11,642 miles) for petrol cars, and electric SUVs cover 17,040km (10,587 miles) against 20,830km (12,945 miles) for petrol SUVs. Even Tesla vehicles, used more than other BEVs, are driven less than petrol cars. Plug-in hybrids and hybrid vehicles have a similar usage rate to petrol vehicles. Researchers also note that manufacturing EVs incurs higher emissions than petrol vehicles, and if EVs aren't used extensively, it could prolong the time to offset these initial emissions through regular use.